SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 2241026 **GROUP NO: 26-10**

PROJECT TITLE: AI-Based Phishing Website Detection using Hybrid Feature Learning

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PROJECT ABSTRACT:

One of the most serious and fastest growing threats on the modern Internet are phishing websites impersonating legitimate web pages to extract sensitive usernames and passwords, banking information and personal information from the user. Customary blacklist-based or rule-based phishing detection techniques fail to detect newly generated or obfuscated phishing domain names. Although machine learning is increasingly being used to reduce phishing threats, existing approaches encounter difficulties, as they are limited by their feature space and use fixed datasets, making them vulnerable to zero-day and adversarial attacks.

The primary goal of this research work is to design and develop an **AI-driven phishing website detection framework through hybrid feature learning** to improve the recognition accuracy, robustness, and generalizability. The proposed framework extracts multi-domain feature representations, including lexical URL embeddings, HTML structural features, host-based metadata, and hyperlink graph features for phishing website classification. Context dependency among URL tokens is captured by a transformer-based encoder, and hyperlink dependency among different hyperlinks is captured by an inter-link dependency-aware module. An attention-based fusion layer is used to combine heterogeneous features for better classification. Furthermore, complementary components such as adversarial training and temporal validation ensure robustness against evolving phishing strategies.

The aim of the framework is to provide scalable, interpretable and fast detection solutions that can be used in browser extensions, enterprise gateways and cloud security services, while overcoming the high false positives and low zero-day detection rates of existing malware detection systems. Research will strengthen digital trust and privacy while improving cybersecurity resilience.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

I. Software:

- Python 3.x
- Jupyter Notebook / VS Code
- Libraries: NumPy, Pandas, Scikit-learn, XGBoost, TensorFlow/PyTorch, BeautifulSoup, Requests, Transformers (Hugging Face), SHAP, Matplotlib
- Dataset: Phishing & Legitimate URLs (PhishTank, OpenPhish, Kaggle / UCI ML Repository)

II. Hardware:

- Intel i5/i7 or equivalent processor
- 8 GB RAM (minimum) – 16 GB recommended
- 100 GB free storage (SSD preferred)

III. Methods / Algorithms:

- Feature Extraction (URL-based, Structural, Hyperlink & Content features)
- URL Tokenization & Semantic Embedding (Transformer-based)
- Feature Scaling & Selection (Correlation Analysis, Recursive Feature Elimination)
- Machine Learning Models:
  - Random Forest
  - Support Vector Machine (SVM)
  - Logistic Regression
  - XGBoost
  - CNN (for deep feature learning)
  - Hybrid / Ensemble Model (Stacking or Soft Voting)
- Model Evaluation: Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

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(3) APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):

.... **Project Supervisor**

.... **Section Coordinator, FRP**

.... **FRP Coordinator**